

EXPENSE TRACKER APPLICATION

PYTHON PROGRAMMING INTERNSHIP – TASK 2

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Internship	MAINCRAFTS TECHNOLOGIES
Domain	Python Programming
Task	Task - 2
Project	Expense Tracker using Python

Project Overview:

In this task, I developed a simple Expense Tracker application using Python. The application allows users to add expenses, store records using CSV files, view saved expenses, and calculate total expenses efficiently.

Objectives of Task-2

- Learn Python file handling using CSV files.
- Understand the use of functions and loops in Python.
- Create a menu-driven command-line application.
- Store and retrieve user data effectively.
- Calculate and manage expense records.

Features Implemented

- Add new expenses with description and amount.
- View all saved expenses from the CSV file.
- Calculate total expenses automatically.
- Use functions for better code organization.
- Store data permanently using **expenses.csv**.

Technologies Used

- Python Programming Language
- CSV Module
- File Handling
- Functions and Loops
- Command-Line Interface (CLI)

Learning Outcomes

Through this task, I improved my understanding of:

- Python functions and modular programming.
- Reading and writing CSV files.
- Building command-line applications.
- Managing user inputs and calculations.
- Creating beginner-level real-world projects.

Conclusion

This task provided practical experience in building a Python-based Expense Tracker application. It strengthened my programming fundamentals and improved my understanding of file handling, functions, and user interaction in Python.

Python Source Code

```
import csv
import os

FILE_NAME = "expenses.csv"

# Function to add expense
def add_expense():
    description = input("Enter expense description: ")

    try:
        amount = float(input("Enter amount: █"))

        with open(FILE_NAME, "a", newline="") as file:
            writer = csv.writer(file)
            writer.writerow([description, amount])

        print("Expense added successfully!\n")

    except ValueError:
        print("Invalid amount! Please enter numbers only.\n")

# Function to view all expenses
def view_expenses():
    if not os.path.exists(FILE_NAME):
        print("No expenses found.\n")
        return

    print("\n----- All Expenses -----")

    with open(FILE_NAME, "r") as file:
        reader = csv.reader(file)

        empty = True

        for row in reader:
            empty = False
            print(f"Item: {row[0]} | Amount: █{row[1]}")

        if empty:
            print("No expenses recorded.")

    print()

# Function to calculate total expenses
def total_expenses():
    if not os.path.exists(FILE_NAME):
        print("No expenses found.\n")
        return

    total = 0

    with open(FILE_NAME, "r") as file:
        reader = csv.reader(file)

        for row in reader:
            total += float(row[1])

    print(f"\nTotal Expenses = █{total}\n")

# Main menu
while True:

    print("===== Expense Tracker =====")
    print("1. Add Expense")
    print("2. View Expenses")
    print("3. View Total Expense")
```

```
print("4. Exit")

choice = input("Enter your choice: ")

if choice == "1":
    add_expense()

elif choice == "2":
    view_expenses()

elif choice == "3":
    total_expenses()

elif choice == "4":
    print("Exiting Expense Tracker...")
    break

else:
    print("Invalid choice! Please try again.\n")
```